

6CCP3131 Third Year Project

Modelling disease dynamics with the Markov Chain Monte Carlo simulation of the Ising Lattice

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MODELLING DISEASE DYNAMICS WITH THE MARKOV CHAIN MONTE CARLO SIMULATION OF THE ISING LATTICE

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ABSTRACT

The project aims to demonstrate the 2-dimensional Ising Lattice's utility in simulating disease diffusion within a closed population, reproducing the findings of Serra, Nicola et al.'s work, "Diffusion modelling of COVID-19 under lockdown conditions" [1]. The report provides a comprehensive theoretical background covering compartmental epidemiology modelling, the Ising Lattice, and computational methodologies for solving the Ising Model. Utilizing the Markov Chain Monte Carlo (MCMC) method, particularly the Metropolis-Hastings algorithm, the study calculates the Ising Lattice's magnetization and energy, with a critical temperature estimate falling within the range of $2.2 < k_B T_C < 2.4$, consistent with theoretical expectations. Furthermore, an analogy is established between the $+1/-1$ spin states in the Ising Lattice and the infected/healthy individuals within a population, and the MCMC is used to simulate disease dynamics. The report argues for and against of such analogy from the perspective of statistics and the physical origins of the Ising Lattice. Additionally, the project extends the MCMC method to account for the duration of an individual's contagiousness or immunity after recovery, enhancing the model's applicability to real-world scenarios. Qualitative comparisons are drawn between Ising Lattice simulations and epidemiological differential equations, offering insights into potential mathematical analogies between the two methods, and the influence of temperature as a degree of randomness on infection occurrence is explored, contributing to a deeper understanding of disease dynamics.

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1. INTRODUCTION

In 1924, Ernst Ising in his PhD thesis tackled the solution of his supervisor's, Wilhelm Lenz's statistical model that described the magnetic behaviour of ferromagnetic materials. [2] Ising demonstrated that a one-dimensional ferromagnetic lattice does not undergo phase transition, and

predicted that higher dimensional lattices would exhibit the same behaviour, dismissing the significance of his model. Although his solution was correct, his predictions about the higher dimensional lattices and the importance of his model were inadequate. In 1944, Onsager analytically extended Ising's research to an infinite two-dimensional square lattice, showing that higher dimensional lattices do

exhibit a second-order phase transition at a critical temperature. This event marked the start of the importance of the so-called Ising Model [3]. Over the past 100 years, scientists found many other applications for the Ising Model due to its critical behaviour, for example analysing the behaviour of economic markets [4] [5], investigating anomalies in IT networks [6], modelling psycho-political phenomena [7], or describing the cloud patterning formation [8].

Due to the COVID-19 outbreak, there has been a recent interest in looking into new mathematical approaches to characterise disease dynamics, predict potential outbreaks, and analyse lockdown conditions, for example using computational fluid-dynamics [9], autoregressive models [10] and linear regression [11, 7], besides the well established, SIR (Susceptible-Infected-Recovered) epidemiology model [12] [13]. The binary nature of the Ising Lattice gave inspiration to the research community to apply it also for epidemiology modelling, drawing either a mathematical [14] or a qualitative analogy [15] [16] [17] [1] between the +/- states of the spins within the lattice, and the healthy/infected individuals within a closed population. A notable publication is Serra, Nicola et al's work, "Diffusion modelling of COVID-19 under lockdown" [1]. They used a modified version of the 2D Ising model with an external field of zero to simulate the diffusion of COVID-19 at various temperatures, employing the Markov Chain Monte Carlo (MCMC) method with a modified transition probability.

This report gives a theoretical background to the compartmental epidemiology models, the Ising Lattice, the analogy between them, and the MCMC procedure. It approximates the solution of the two-dimensional Ising Lattice using the Metropolis-Hastings algorithm. Additionally, it aims to outline the methodology and reproduce the results of the paper of Serra, Nicola et al, "Diffusion modelling of COVID-19 under lockdown" [1], arguing for and against such analogy, and suggesting improvements. Lastly, the report introduces an extension to the MCMC solution of the Ising Lattice to also account for recovered individuals that have developed immunity and the amount of time an individual stays infected.

2. BACKGROUND THEORY

2.1. Epidemiology Models

Compartmental epidemiology models are mathematically simple yet effective techniques to model the spread of infectious diseases within a population. [18] The simplest compartmental model

is the SI model, which segments the population into two states: the number of individuals susceptible to catching the disease (S) and the number of infected individuals (I). The model assumes a constant population over time, $S(t) + I(t) = N$ (no births or deaths) and that when an individual catches the disease, it stays infected. The time evolution of the two compartments can be captured by the deterministic differential equations:

$$\begin{cases} \frac{dS}{dt} = -\beta S(t)I(t) + \gamma I(t) \\ \frac{dI}{dt} = \beta S(t)I(t) - \gamma I(t) \end{cases} \quad (1),$$

where β is the rate of infection and an infected individual can recover and become susceptible again with rate γ . [18]. Given that there are some infected individuals present in the population initially, ($I(0) \neq 0$), the infection will spread. Figure 1. shows an example solution of the SIS model, infecting 1% of the population initially.



Figure 1: An example solution of the SIS model (Equation 1), setting the infection rate to $\beta=0.3$, the recovery rate to $\gamma=0.1$, and using the initial conditions $I(0) = 0.01$ and $S(0) = 0.99$.

An improvement of the SIS-model is the introduction of the 'Recovered' population segment, which entails individuals that recovered after infection, and developed an immunity to the disease (SIR-model) The rate equations for such model are given by Equation 2:

$$\begin{cases} \frac{dS}{dt} = -\beta S(t)I(t) \\ \frac{dI}{dt} = \beta S(t)I(t) - \gamma I(t) \\ \frac{dR}{dt} = \gamma I(t) \end{cases} \quad (2).$$

The example solution of the SIR model is plotted in Figure 2., using the same rates and initial conditions as in Figure 1.

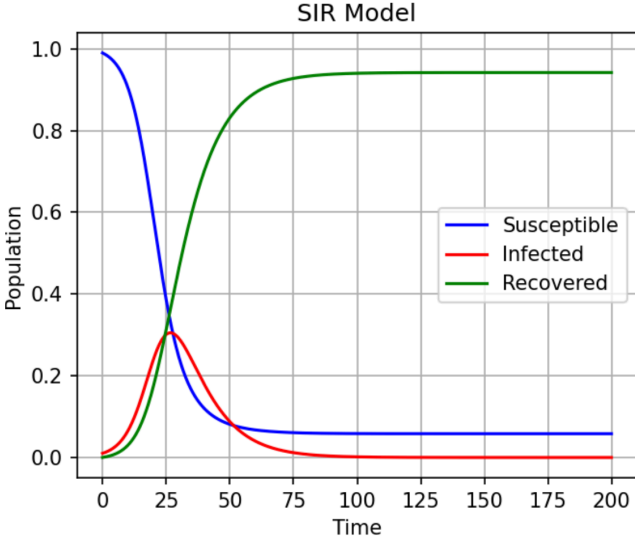


Figure 2: An example solution of the SIR-model (Equation 2), setting the infection rate to $\beta=0.3$, the recovery rate to $\gamma=0$, and using the initial conditions $I(0) = 0.01$, $S(0) = 0.99$ and $R(0) = 0$.

If individuals who have recovered from an infection can lose immunity and become susceptible again after a certain period of time, the SIRS model is employed. In this model, the parameter ζ represents the rate at which recovered individuals transition back to the susceptible state and can become infected again. The dynamics of this model are captured by the set of differential equations

$$\begin{cases} \frac{dS}{dt} = -\beta S(t)I(t) + \zeta R(t) \\ \frac{dI}{dt} = \beta S(t)I(t) - \gamma I(t) \\ \frac{dR}{dt} = \gamma I(t) - \zeta R(t) \end{cases} \quad (3).$$

Figure 3. illustrates an example solution for the SIRS model, using the same initial conditions, infection rate and recovery rate as Figures 1. and 2., and setting $\zeta = \beta$.

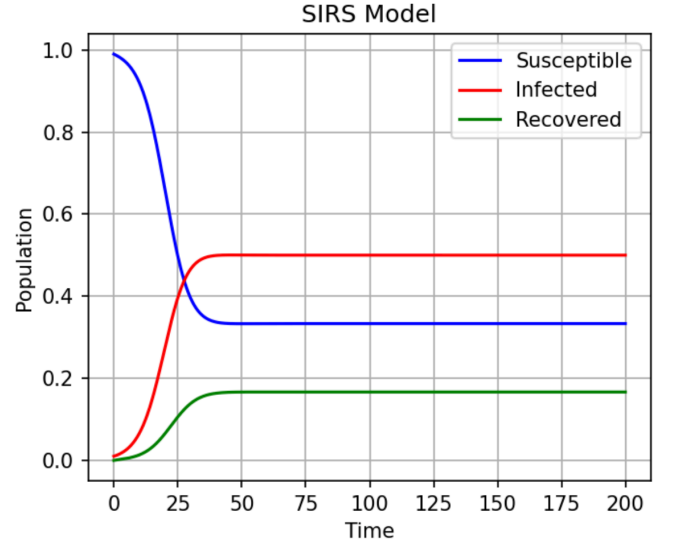


Figure 3: An example solution of the SIRS-model (Equation 2), setting the parameters to $\beta=\zeta=0.3$, $\gamma=0$, and the initial conditions to $I(0) = 0.01$, $S(0) = 0.99$, $R(0) = 0$.

The rate parameters in compartmental models can be adjusted to reflect real-world disease dynamics. However, due to the model's simplicity, only the most crucial factors of disease transmission can be captured, for example, disease type, contact frequency, or environmental factors such as temperature and humidity. For a more nuanced understanding, the 2-dimensional Ising Model offers a solution by incorporating spatial interactions, enabling simulations that account for localized transmission dynamics.

2.2. The Ising Model

In the Ising model, a two-dimensional magnet is defined by a square lattice with N sites. Each site corresponds to a dipole in the magnet, which can take the spin states $\sigma_i = -1$ or $+1$. Using the assumption that the nearest neighbouring spins can interact, the Hamiltonian describing the Ising lattice is given by

$$H = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - h \sum_i \sigma_i \quad (4),$$

where J is the interaction strength between the two neighbouring spins, h is the external magnetic field applied to the system, and $\langle i,j \rangle$ denotes the sum over the nearest neighbours. The first term within the Hamiltonian gives the interaction energy between neighbouring spins, while the second term describes the response of the dipoles to the external field. Using the laws of statistical physics, in a canonical ensemble, the probability of finding a particular microstate of configuration $\{\sigma_i\}$ is given by

$$p(\{\sigma_i\}) = \frac{1}{Z} e^{-H(\{\sigma_i\})/k_B T}, \quad Z = \sum_{\{\sigma_i\}} e^{-H(\{\sigma_i\})/k_B T} \quad (5),$$

where Z is the partition function. This probability gives the insight that when $J > 0$, the interaction energy between the neighbouring spins favour parallel alignment (ferromagnetism), while in the case of $J < 0$, neighbouring spins prefer anti-align (anti-ferromagnetism). The normalised sum of the spin states of the lattice gives the magnetisation of the lattice, M :

$$M = \frac{1}{N} \sum_i \sigma_i \quad (6).$$

2.3. Critical Behaviour

A key feature of ferromagnetic materials ($J > 0$) is that in the absence of an external field ($h = 0$), they can exhibit spontaneous magnetization. This magnetic momentum is temperature-dependent and disappears above the Curie temperature (T_c), which was calculated analytically by Onsager [3]

$$T_c = \frac{2J}{\ln(1 + \sqrt{2}) k_B} \approx \frac{2.27J}{k_B} \quad (7).$$

From a physical perspective, this phase transition can be explained by the temperature dependence of the thermal fluctuations and the interaction energies of the dipoles. At low enough temperatures ($0 < T < T_c$), the thermal energy of the dipoles is much lower than the interaction energy between the neighbouring spins, hence the ferromagnetic interaction between the spins dominates the system, according to Equation 5. Macroscopic-sized clusters form and the system gains spontaneous magnetization. In equilibrium, one cluster takes on a dominant role, determining the sign of the magnetization, and a ferromagnetic, ordered phase occurs.

When $T > T_c$, the thermal energy of the dipoles are larger than the interaction energy between the neighbouring spins, and this thermal fluctuation breaks the clusters. The system loses its original magnetization, and the system transitions to a disordered, paramagnetic phase with second-order phase transition.

The absolute value of the magnetisation of the lattice, $|M|$ gives insight into how much ordered or disordered the phase is, hence it can be used as an order parameter. Below the critical temperature, $|M|$ tends towards 1, indicating an ordered phase, while above the critical temperature, $|M|$ approaches 0, signifying a disordered phase.

If the Ising Lattice is heated up from $T=0$ to $T > T_c$, the system undergoes the transition from ordered to disordered phase ($|M|=0 \rightarrow |M|=1$), and it traverses its critical point at T_c . As the critical temperature is approached, length diverges, meaning that the fluctuations in the system occur on all length scales, and at T_c the correlation length becomes infinite. This leads to the formation of clusters of various sizes, from small domains to large-scale structures. Therefore, the standard deviation of the mean magnetisation, $\Delta|M|$, is expected to be the largest around the critical temperature. This phenomenon occurs because, during the phase transition, the system has a heightened sensitivity to perturbations in any of its components, hence to the change in temperature as well. [19]

2.4. Monte Carlo Method

Up to date, the Ising Model can only be solved analytically in one or two dimensions. However, with the emergence of numerical techniques, higher dimensional Ising lattices have been successfully solved, with one of the most popular methods up to date being the Monte Carlo method. [20]

To describe the behaviour of the Ising lattice at various temperatures and later use it as an analogy for epidemiology models, the mean magnetisation of the lattice $\langle M \rangle$ was of interest at specific temperatures. However, for a lattice with N sites, one would need to examine all 2^N configurations to determine $\langle M \rangle$. The Monte Carlo (MC) method can assist in numerically approximating the desired physical parameter, by selecting only a subset of configurations based on importance sampling.

2.4.1. Random Sampling

The probability of selecting a particular configuration can be determined in various ways. In random sampling, each configuration is chosen with equal probability. However, in the case of the Ising model, the individual microstates do not occur with the same probability, so the states need to be weighted by their occurrence frequency, according to Equation 5:

$$\langle M \rangle = \frac{1}{Z} \sum_r M_r e^{-E_r/k_B T} \quad (8),$$

where M_r represents the magnetism of the Ising lattice in microstate r , and E_r denotes the energy of the lattice, calculated according to the Hamiltonian defined in Equation 4. Besides temperature, other parameters, such as external field strength and the interaction strength J can influence the weight value through the energy appearing in the Boltzmann factor. The summation is performed over all the possible microstates, indexed by r . This method is

not applicable when the energy distribution of the system is not sharp enough or when rare states occur significantly affecting the expected value of the parameter. However, typically, under certain parameters, the energy exhibits a sharp maximum, so only a very limited set of configurations determines the value of the physical parameter under those factors. It can only be applied for very limited system sizes; even examining a 10×10 lattice entails $2^{100} \simeq 10^{12}$ states, which double with each additional lattice point. [20]

2.4.2. Importance Sampling

In importance sampling, the configurations that significantly influence the average of the measured physical quantity are selected more easily among the measurement points. This can be achieved by adjusting the selection probability precisely to the Boltzmann factor $P_r = \frac{1}{Z} e^{-E_r/k_B T}$, and the estimation of the average value simplifies to:

$$\langle M \rangle = \frac{1}{N} \sum_r M_r \quad (9).$$

If the system has one or more sharp, characteristic energy values, and these states can be easily identified, this method provides a manageable approach even for larger systems.

While importance sampling partly solves the implementation of the measurement, the problem remains with how to choose configurations whose Boltzmann factors, given the known macroscopic quantities, are large enough to significantly influence the value of the measured quantity. It is necessary to find not only the first but also the other measurement points, preferably in such a way that they essentially represent different configuration groups and are independent of the previous ones. This is where the Markov chain comes into the picture.

2.5. Markov Chains

A Markov Chain denotes a sequence of configurations $C_0, C_1, \dots, C_n$, where each configuration is traversed in such a way that the previous, and only the immediate previous configuration becomes the starting point for selecting the next one. In this chain, the probability of transitioning from state $C_i = k$ to state $C_{i+1} = l$ is given by $P(k \rightarrow l)$. These probabilities must satisfy the condition such that $\sum_l P(k \rightarrow l) = 1$ for all states k . The observed quantity of interest, in our case $\langle M \rangle$, can be measured over the configurations C_i of the chain. For the observed quantity to tend on average toward the desired average,

$$\langle M \rangle_C \rightarrow \langle M \rangle + O(n^{1/2}) \quad (10),$$

each transition must satisfy two conditions. Firstly, the transition probabilities have to adhere to the principle of detailed balance,

$$\frac{P(k \rightarrow l)}{P(l \rightarrow k)} = \frac{\pi(l)}{\pi(k)} \quad (11),$$

where $\pi(l)$ and $\pi(k)$ are the probabilities of being in states l and k respectively according to the target distribution. The second condition to satisfy the convergence of the Markov Chain is the Ergodicity, which states that from all states l there must be a non-zero probability to transition to any state k , hence from any initial state all the possible configurations should be able to be explored given sufficient time. [20]

2.6. The Metropolis-Hastings Algorithm

A type of Markov Chain Monte Carlo method is the Metropolis-Hastings Algorithm [21], which operates by proposing changes to a given configuration. The proposal is subsequently accepted or rejected with a certain probability, with accepted proposals becoming the succeeding element in the Markov Chain. The formulation of the transition probability ensures that sampled configurations adhere to the specified target distribution. This described procedure constitutes a fundamental unit in the MCMC algorithm, known as a Monte Carlo step.

The Metropolis-Hastings algorithm is a spin-flip algorithm, where the proposed configuration is generated by flipping a single spin. Another example of the MCMC methods are the cluster-flip algorithms, for example, the Wolff-algorithm [22], where one or more entire homogeneous regions are flipped between two steps. While the cluster-flip algorithms are more computationally efficient, this report implements the Metropolis-Hastings Algorithm due to its suitability for the epidemiological analogy.

The implementation of the algorithm proceeds as follows: initially, the Ising lattice is in a random configuration, denoted as C_i , at a specific temperature T . Then, a spin is randomly selected and its sign is flipped, marking a Monte Carlo step. To ensure the importance sampling described in Section 2.4.2, this new configuration is automatically accepted if the Monte Carlo step results in lower energy, which is considered energetically favourable. If the step is energetically unfavourable, it is accepted with a probability determined by the Boltzmann weight. If state k is energetically more favourable, then the transition probabilities are:

$$P(l \rightarrow k) = e^{-(E_l - E_k)/k_B T}, \quad P(k \rightarrow l) = 1 \quad (12),$$

where E_l and E_k are the energies of the Ising Lattice in state l and k , respectively. By accepting energetically less favourable configurations with a non-zero probability, the algorithm allows for the exploration of the whole solution space. Thus, the likelihood of converging to a metastable minimum instead of the desired global minimum can be reduced by employing a sufficiently large number of Monte Carlo steps. Moreover, the probabilities enable all spins in the lattice to be flipped, ensuring an ergodic update scheme. The probabilities also fulfil the detailed balance condition (Eq 11.):

$$\frac{P(k \rightarrow l)}{P(l \rightarrow k)} = \frac{\pi(l)}{\pi(k)} = \frac{e^{-E_l/k_B T}}{e^{-E_k/k_B T}} = e^{-(E_l - E_k)/k_B T} \quad (13).$$

2.7. The Analogy between the SI epidemiology model and the Ising lattice

To make the analogy between the SI model and the 2-dimensional Ising lattice, each spin can be corresponded to an individual within a closed population. A spin state -1 can represent a healthy individual, while a spin state $+1$ an infected one. The interaction with the closest neighbours corresponds to contacts in a population. It has to be noted that this analogy assumes that an individual has exactly 4 contacts, which in real life is not always the case.

The external magnetisation is switched off ($h=0$) and the magnetisation of the lattice, M gives the overall health status of the population. In this analogy, an $M = -1$ phase means that the population is healthy. The disordered phase, $M = 0$, was interpreted in the paper of SN [1] as a state where the infectious disease has penetrated the population (the disease has diffused), with each individual having a 50-50% probability of being infected, hence epidemics occurred. To simulate an initial healthy population, the simulations will be run using the phase $M = -1$ as the initial configuration. Sufficiently above T_c , $M = 0$ can be expected, and sufficiently below T_c , $M = -1$. While the probability of observing $M = +1$ below T_c with the initial configuration $M = -1$ is non-zero, it is negligibly small, making it practically impossible. Therefore, the state $M = +1$ is not used in this analogy, as it will not appear in the simulation.

Setting the interaction strength to $J > 0$ ensures parallel alignment, meaning that when neighbouring spins are infected ($+1$), the likelihood of a healthy spin (-1) getting infected also increases. This mirrors the behaviour of epidemics, where the spread of infection is influenced by the infected individuals around a susceptible person. However, this analogy suggests that the probability of recovery (flipping the spin from -1 to $+1$) is also influenced by the health status of neighbouring spins. This aspect diverges

from real disease dynamics, where the recovery depends on factors like the type of disease, duration of infection, and the individual's health conditions.

In SN's paper [1], the energy of the lattice has been corresponded to the mobility of the population under lockdown conditions. This analogy can be argued by examining the Hamiltonian of a single, healthy spin, $\sigma_i = -1$ without an external field:

$$H_i = -J\sigma_i \sum_{(j)}^N \sigma_j = J \sum_{(j)}^N \sigma_j \quad (14).$$

It can be deduced from the equation above that a healthy spin ($\sigma_i = -1$) has increasing energy as the number of infected neighbours increases. Consequently, during the Metropolis-Hastings procedure, it becomes increasingly favourable for the spin to flip, as dictated by Equation 12. In the context of epidemiology, this can be corresponded that if an individual has greater mobility (higher energy), it suggests that they have more contacts with infected individuals, thus increasing the likelihood of infection (flipping the spin from -1 to $+1$). This assumption seems reasonable given the dynamics of disease transmission.

The transition probability from a healthy state to an infected state also depends on the temperature (Equation 12). SN [1] interpreted the temperature as the actual temperature of the environment in which the disease spreads. In their model, they considered that experiments showed that the probability of infection in the cases of respiratory diseases varies with temperature according to the Gaussian function:

$$\lambda(T) = ae^{-\frac{1}{2}(\frac{\mu-T}{\sigma})^2}, \quad a = 1; \mu = 10; \sigma = 10 \quad (15).$$

Figure 4. illustrates the Gaussian function shown in Equation 15.

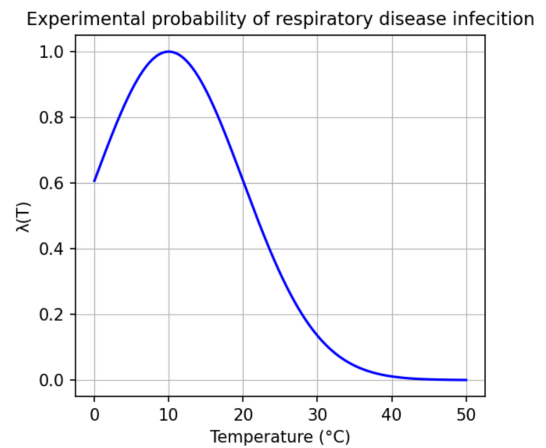


Figure 4: The probability of respiratory disease infection as a function of temperature, $\lambda(T)$, given by Eq 15. The function was obtained based on experimental results. [1]

Figure 4. shows that the transmission of the disease is most probable between 0 and 20 °C, it peaks at 10 °C and significantly decreases above 30 °C. To account for such seasonality of the respiratory diseases, SN [1] weighted the transition probability with this statistical parameter:

$$P(l \rightarrow k) = e^{-(E_l - E_k)/k_B T \lambda(T)}, \quad P(k \rightarrow l) = 1 \quad (16).$$

3. METHODOLOGY

3.1. The standard Metropolis-Hastings procedure

For benchmarking purposes, firstly a standard, N by N Ising Lattice has been solved with the Metropolis-Hastings algorithm using sizes N=10 and N=20 with periodic boundary conditions. The external magnetic field has been set to zero and the interaction strength to $J = 1$ [J]. For the Ising Model, the natural unit of energy and temperature is [J]. To simplify further calculations, [J] is set to 1 throughout the paper.

Firstly, a random configuration of the Ising Lattice has been generated. Then, using temperature T_i , the Monte Carlo steps described in Section 2.6 have been repeated N^2 times within a function called a “lattice sweep”. This way it has been ensured that within a lattice sweep, each spin had an equal probability of being chosen. Hence, the transition probability had been modified to

$$\tilde{P}(l \rightarrow k) = \frac{1}{N^2} P(l \rightarrow k) \quad (17),$$

where $P(k \rightarrow l)$ became the probability of transition from l to k provided that state l has been chosen. This modification did not violate the detailed balance condition, as plugging it back into Equation 11, the additional factor of $1/N^2$ vanishes:

$$\frac{P(k \rightarrow l)}{P(l \rightarrow k)} = \frac{\tilde{P}(k \rightarrow l)}{\tilde{P}(l \rightarrow k)} \quad (18).$$

Each lattice sweep procedure has generated a final microstate, and the absolute value of the magnetisation of the lattice has been stored, denoted as M_r . Then, the lattice sweep procedure has been iterated over n_{sweeps} times, using the final configuration of the previous sweep as the initial configuration of the next lattice sweep. Ensuring that n_{sweep} is sufficiently large enough, the system reaches an equilibrium according to Equation 10, where the microstates in equilibrium are distributed according to the Boltzmann weight. The physical parameter of interest of the simulation, $\langle M \rangle$ has been estimated by taking samples from the microstates in equilibrium and averaging their values:

$$\langle M \rangle_{\text{sample } T_i} = \frac{1}{n_{\text{samples}}} \sum_n M_{\text{req}} \quad (19),$$

where n_{samples} are the number of samples taken, T_i denotes the temperature used for the simulation, and M_{req} are the magnetisation of the microstates in equilibrium.

The steps described above have been repeated for temperatures ranging from $k_B T=0$ to $k_B T=5$, with increments of $\Delta k_B T = 0.1$. For the first simulation, the temperature has been set to the highest value, $k_B T_0 = 5$, with an initial random configuration, C_0 . For the consecutive simulations, the temperatures $k_B T_{i+1} = k_B T_i - 0.1$ have been used, using the final configuration of the subsequent simulation as the starting configuration for the consequent. This method uses the physical knowledge of the Ising system that at high temperatures a disordered phase can be expected, and with decreasing temperatures, the lattice has a second-order, smooth phase transition. The simulation would also work if an arbitrary initial state were chosen for each temperature, however, it would take more computational time to reach equilibrium. [20]

To produce statistically independent values, the procedure described above has been repeated $n=5$ times using different random seed generators for the initial configurations, and the final reported mean magnetisation for each temperature, $\langle M \rangle_{T_i}$, has been obtained by:

$$\langle M \rangle_{T_i} = \frac{1}{n} \sum \langle M \rangle_{\text{sample } T_i} \quad (20).$$

The error for the final obtained result can be calculated using the equation

$$\Delta \langle M \rangle_{T_i} = \frac{\sigma}{\sqrt{n}} \quad (21).$$

where σ is the standard deviation of the $\langle M \rangle_{T_i}$ values.

The number of iterations of the lattice sweep has been set to $n_{\text{sweeps}} = 2000$, and equilibrium samples were taken after the first 1000 sweeps to allow enough time for the equilibration.

The critical temperature has been estimated by approximating the temperature where $\Delta \langle M \rangle$ showed to be the largest on average for all lattice sizes (see Section 2.3). The mean energy of the lattice, $\langle E \rangle$ has been also calculated using the same method as for, $\langle M \rangle$.

3.2. Simulation of the Diffusion of COVID-19

To reproduce the results of SN's paper [1], the same procedure described in Section 3.1 has been repeated with the following modifications:

1. The transition probability has been modified according to Equation 16.
2. The starting configuration at each temperature has been set to $M = -1$, representing a healthy population as the initial configuration
3. The simulation has been run at temperatures ranging between $T=0-50^\circ\text{C}$ with increments of 1°C , as the respiratory diseases spread with a significant probability between these values (see Figure 4), converting the units of $k_B T$, [J] to $[\text{K}]$.

In the paper, a 20 by 20 lattice size has been used, with $J = 2$, $h = 0$. SN has generated 30 lattice sweeps for each temperature and the sample $M_{r,eq}$ values were calculated from the last 20 configurations. To reproduce the results, the same parameters have been used. To see how the system behaves with increased precision, the simulation has also been run with 2000 lattice sweeps, calculating the equilibrium values from the last 1000 sweeps.

SN in their paper does not mention the condition for detailed balance. The only way to obey this condition and to have the Markov Chain converge to the desired value is to assume that the target distribution of state k , $\pi(k)$ also gets modified according to Equation 13:

$$\frac{P(k \rightarrow l)}{P(l \rightarrow k)} = \frac{\pi(l)}{\pi(k)} = \frac{e^{-E_l/k_B T \lambda(T)}}{e^{-E_k/k_B T \lambda(T)}} = e^{-(E_l - E_k)/k_B T \lambda(T)} \quad (22).$$

3.3. Extending the MCMC procedure to account for the duration of contagiousness and immunity

The analogy drawn between the Ising and SI models in Section 2.7 suggests that an infected spin can recover at any Monte Carlo step with a probability dictated by Boltzmann's distribution. However, from an epidemiological perspective, this assumption lacks realism. In reality, an infected individual should not have the ability to recover at any arbitrary time but rather should have a recovery probability contingent upon the duration of the infection. Moreover, the application of the Boltzmann weight implies that the likelihood of recovery is influenced by the health status of neighbouring spins, as discussed in Section 2.7.

To account for these observations, a new extension has been developed. Similarly to SN's method, at the beginning of each simulation, the initial

configuration is set to $M = -1$, indicating that initially all spins are healthy (susceptible). Each Monte Carlo step is interpreted as a healthy spin getting in contact with the disease and getting infected according to the Boltzmann probability. Therefore, a sweep has been made analogous to a timestep, meaning that timestep, each susceptible individual can catch the disease with the Boltzmann probability. The new idea is that if a healthy spin gets infected with the Boltzmann probability during a sweep, it would stay infected until a certain amount of lattice sweeps with a 100% probability. After this time has passed, the spin would recover and become susceptible again with a 100% probability, simulating a more realistic evolution of an infection. A further step is to also account for immunity: after the spin has recovered, it would stay in a healthy state with a 100% probability until a certain amount of time. To implement these extensions, two new parameters have been introduced into the model:

- **Infectious time I_i :** The number of sweeps until the spin stays infectious ($\sigma_i = +1$), after it got infected ($\sigma_i = -1 \rightarrow \sigma_i = +1$),
- **Recovery time R_i :** The number of sweeps until the spin stays healthy ($\sigma_i = -1$) with a 100% probability after it recovered from infection ($\sigma_i = +1 \rightarrow \sigma_i = -1$)

A potential evolution of a spin during the sweeps is shown in Table 1, setting $I_i = 3$ and $R_i = 2$. The spin has also been colour-coded based on three categories:

- **Susceptible, S:** the spin is healthy ($\sigma_i = -1$) but it can get infected with the Boltzmann probability in the next lattice sweep
- **Infected, I:** the spin is infected ($\sigma_i = +1$)
- **Recovered, R:** the spin is healthy ($\sigma_i = -1$) with immunity; it cannot get infected in the next lattice sweep.

The SIR labelling corresponds to the compartments of the epidemiology models discussed in Section 2.1.

Spin's state in the Ising Lattice	Probability of spin-flip during a Monte Carlo Step
-1	Boltzmann weight
-1	
-1	
+1	0%
+1	
+1	
-1	100%
-1	0%
-1	0%
-1	Boltzmann weight
-1	

Susceptible	Infected	Recovered
-------------	----------	-----------

Table 1.: A table showing a potential evolution of the spin status and the SIR status (Susceptible-Infected-Recovered) of a single spin during 9 Monte Carlo steps. Spin -1 indicates a healthy state, while spin +1 denotes an infected status.

The challenge with this modification of the Metropolis-Hastings algorithm is that this method interprets the Monte Carlo calculation as a dynamical calculation. Additionally, by introducing history into the transition probability, the successive configurations are no longer members of a Markov chain. Hence, justifying the application of Boltzmann weight for transition probability solely on the rationale outlined in Section 2.4 is no longer viable. The report opted for its utilization primarily on heuristic grounds. Firstly, a probability needs to be normalizable, for which an exponential form is suitable. Secondly, the transition probability from a healthy to an infected spin has to depend on the spin's energy, as it is connected to the number of infected neighbours as discussed in Section 2.7. Thirdly, it can be argued that the temperature in the denominator of the exponent controls a degree of randomness, governing the probability of a state's occurrence. Thus, the temperature serves as a parameter that influences the probability of infection, termed later in the paper as the "effective temperature," denoted as T_{eff} . Taking all considerations into account, the exponential form of the Boltzmann weight can be regarded as a universal weighting factor.

The algorithmic implementation of the MCMC extension follows these steps:

1. Set the effective temperature, T_{eff} to a specific value.
2. Initiate the Ising Lattice with the configuration where all spins are in the healthy state, $\sigma_i = -1$, representing an initial, healthy population.
3. Initiate a matrix $\{\delta_i\}$, that keeps track of the Susceptible-Infected-Recovered status of the spins. Initially, assign to all spins susceptible states: $\delta_i = S$.
4. Sweep the lattice: Using the SIR status of the spins stored in $\{\delta_i\}$, iterate over all the susceptible spins in a random sequence. For each spin, calculate its energy, E_i , calculate its energy with its spin flipped, E_k , and keep the spin flipped with the probability Equation 12 (accept or reject a Monte Carlo step). Store the indices of the flipped spins. Note that the flipped spins have made the transition from a healthy to an infected state, hence they are the newly infected spins after the lattice sweep.
5. Update the SIR status of the newly infected spins σ_i within the $\{\delta_i\}$ matrix (change $\delta_i = S$ to $\delta_i = I$)
6. Repeat steps 2) and 5) n_{sweep} times
7. If I_t number of sweeps has passed after spin σ_i got infected, after step 4), recover the spin by changing its health status from $\sigma_i = +1$ to $\sigma_i = -1$. Keep track of this SIR state change within the $\{\delta_i\}$ matrix (change $\delta_i = I$ to $\delta_i = R$)
8. If R_t number of sweeps have passed after spin σ_i got recovered, after step 4), make the susceptible again by changing its SIR state from $\delta_i = R$ to $\delta_i = S$ in the $\{\delta_i\}$ matrix. This step allows the algorithm to asses the newly recovered spins at the next lattice sweep, and flip their spin to the infected state (+1) with the Boltzmann probability.

The algorithm described above has been repeated setting the effective temperatures to different values, ranging between $k_B T_{\text{eff}} = 0$ and 8. During the simulation, the magnetisation of the lattice and the number of S/I/R spins have been stored after each lattice sweep, at each temperature. The Python code for how the steps of the algorithm have been implemented is included in the appendix.

4. RESULTS

4.1. The standard Metropolis-Hastings procedure

The results for the standard MCMC procedure are presented in Figure 5 below. The first plot depicts the calculated mean magnetisation at various temperatures, while the second plot shows the mean lattice energy. It can be seen in the figure that the standard deviation increases around the phase transition.

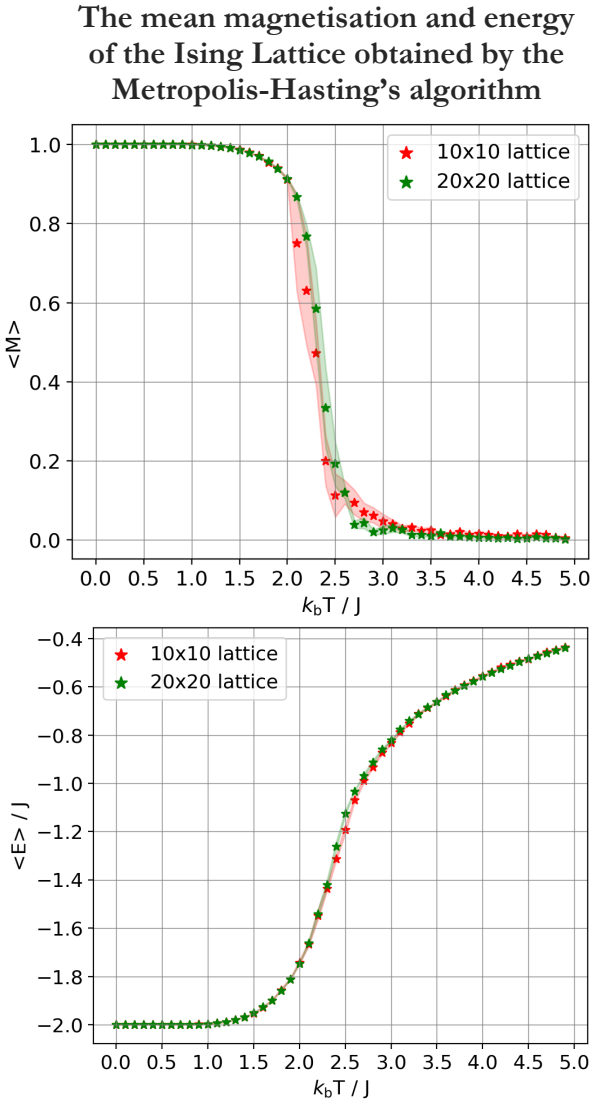


Figure 5: Plots showing the absolute value of the average magnetisation, $\langle |M| \rangle$ (at the top) and the lattice energy $\langle E \rangle$ (at the bottom) of the Ising lattice with sizes 10x10 and 20x20 at various temperatures. The results were calculated with the MCMC Metropolis-Hastings algorithm. The shaded areas show the errors, $\Delta \langle |M| \rangle$ and $\Delta \langle E \rangle$, obtained by using Equation 21.

To estimate the critical region, standard deviations are plotted for temperatures close to this region, shown in Figure 6.

The standard deviation of the mean magnetisation and energy of the Ising Lattice, obtained by the Metropolis-Hasting's algorithm

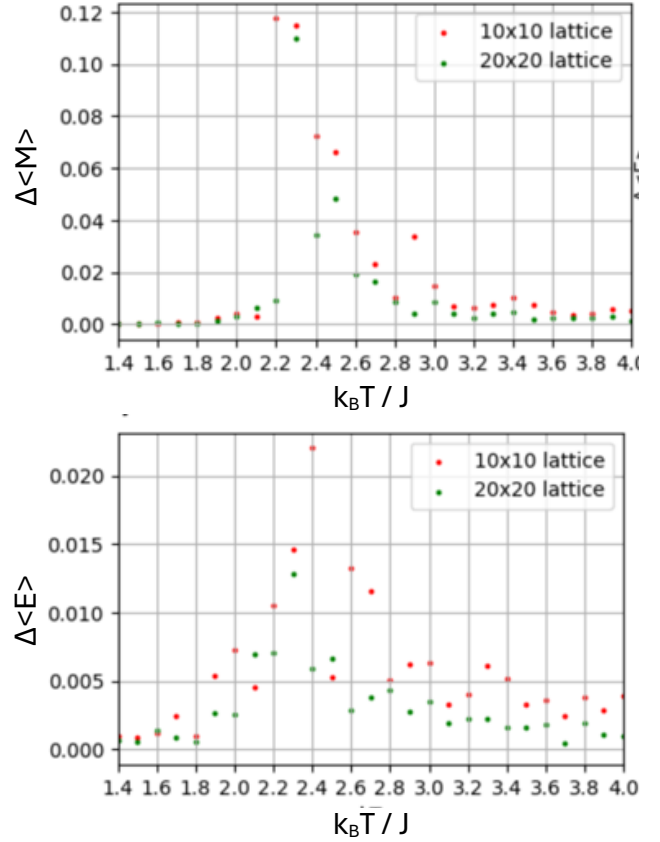


Figure 6: The standard deviation of the mean magnetism and the lattice energy obtained by the MCMC Metropolis-Hastings algorithm, $\Delta \langle |M| \rangle$ and $\Delta \langle E \rangle$ plotted between the temperatures where they showed the largest values.

Figure 6. shows that the most significant $\Delta \langle |M| \rangle$ occurs within the temperature range of $k_B T = 2.1$ to 2.4 , corresponding to the critical temperature region, as discussed in Section 2.3. Therefore, it can be inferred that the critical temperature lies within the range of $2.1 < k_B T_c < 2.4$

4.2. Simulation of the Diffusion of COVID-19

The results obtained by SN [1] are shown in Figure 7 to provide a comparison. The top panel illustrates the mean magnetisation of the lattice at various temperatures, interpreted as the diffusion of COVID-19. The top figure depicts the relationship between the mean lattice magnetisation and energy at each temperature, where the energy has been interpreted by SN as the mobility of the population. The results show that below ~ 2.5 °C, the lattice is in the ordered state, $\langle M \rangle = -1$, indicating a healthy population. Between ~ 2.5 - 10 °C, an increase in

$\langle M \rangle$ can be observed, approaching the disordered phase $\langle M \rangle = 0$, analogous to an epidemic phase. Between ~ 10 - 26°C , the disordered (epidemic) phase persists, and after 35°C , the system transitions back to the ordered, healthy state. It is interesting to note that SN's results show two phase transitions, while the standard Ising model has only one.

SN's results of simulating the diffusion of COVID-19 with the Ising Model

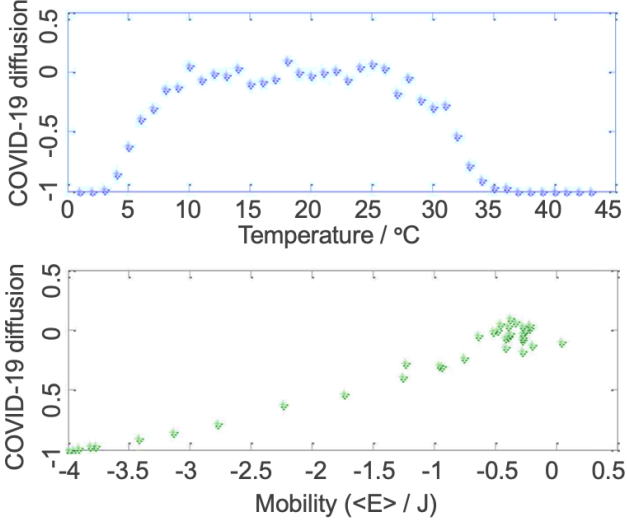


Figure 7: The results of SN's paper [1]. The diffusion of COVID-19 within a closed population corresponds to the mean magnetisation of a 20 by 20 Ising Lattice, while the mean energy of the lattice is interpreted as the mobility of the population. The interaction strength $J=2$ and external magnetic field $h=0$ were used. The results were obtained by the MCMC Metropolis-Algorithm, using a total of 1.2×10^4 MC steps.

The report's attempt to reproduce SN's results is shown in Figure 8. It can be seen that the overall shape of the functions aligns with SN's results (Figure 7), however, sharper phase transitions were obtained.

Reproducing SN's results

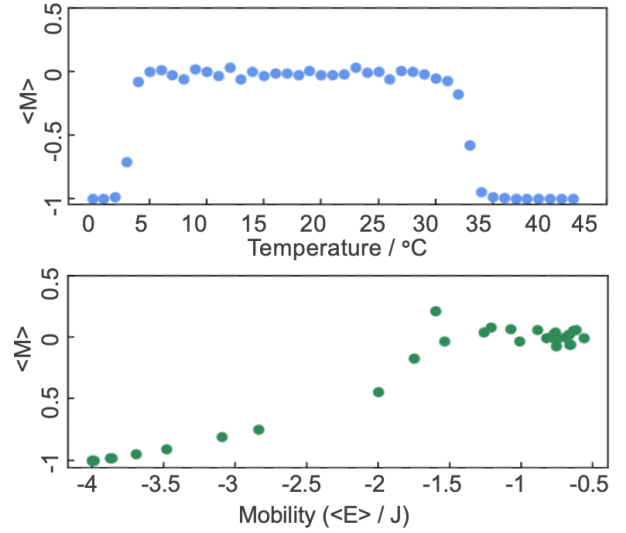


Figure 8: The reproduced results of SN's methodology, using the same parameters and techniques as for Figure 7.

The report has attempted to enhance the results by increasing the Monte Carlo steps from 1.2×10^4 to 8×10^{16} . The new results are displayed in Figure 9, predicting a COVID-19 outbreak between 5 - 30°C .

Improving SN's results

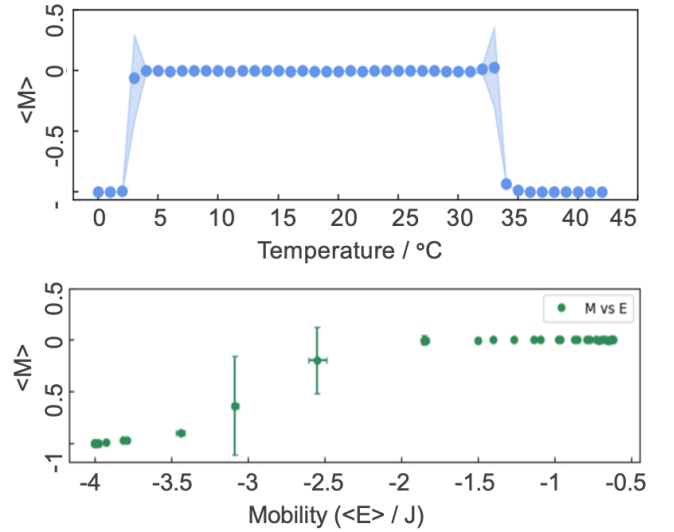


Figure 9: Enhanced results of SN's methodology keeping the Ising lattice size (20 by 20), the interaction parameter ($J=2$) and the external magnetic field ($h=0$) constant, while significantly increasing the Monte Carlo steps from 1.2×10^4 to 8×10^{16} . The simulation was repeated 10 times, and the standard deviation of the results, depicted by shaded areas on the plot, was obtained by using Equation 21.

4.3. Extending the MCMC procedure to account for the duration of contagiousness and immunity

Firstly, the extension was simplified to include only two compartments: the Susceptible and Infected states, simulating the SIS model. Figure 10 presents the result of these simulations at various effective temperatures. The left panels provide an overview of the proportion of spins within the Ising Lattice that are categorized into the Susceptible-Infected state, while the plots on the right showcase the magnetization of the lattice at each sweep iteration.

The simulations were run at effective temperatures ranging between 0 and $8 k_B T$. In Figure 10 three temperatures are plotted, showing the effective temperature range at which the number of infected spins approaches or overtakes the number of susceptible spins.

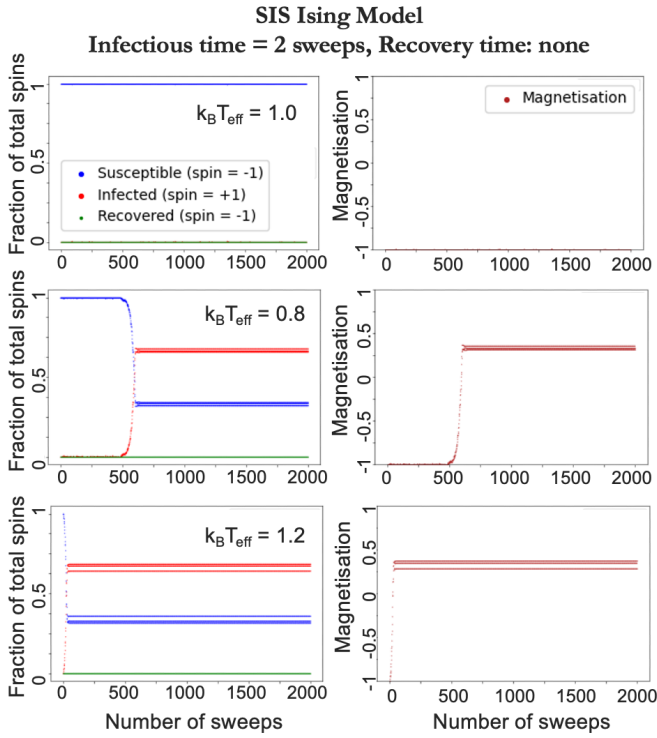


Figure 10: Plots illustrating the impact of incorporating infectious time into the epidemiological simulation for the Ising Model (see Section 3.3). The left-hand figures depict the fraction of Susceptible/Infected spins at each lattice sweep, while the right-hand figures illustrate the magnetization of the lattice throughout the sweeps. The three rows showcase three separate simulations, each using a different effective temperature.

Figure 11 presents the results of the simulation with the introduction of the recovery compartment to the model, where the immunity time was set to infinite, preventing the recovered spins from becoming susceptible again (SIR model).

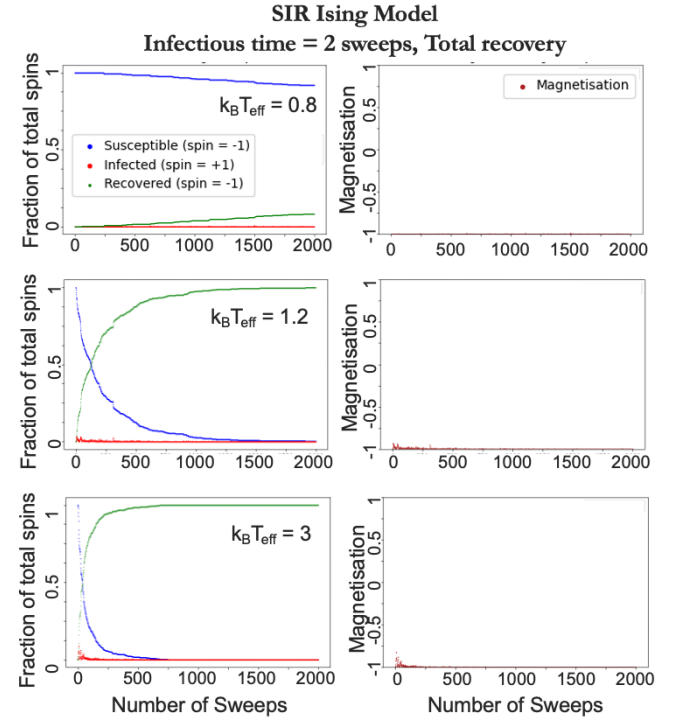


Figure 11: Plots illustrating the impact of incorporating infectious time and recovery into the epidemiological simulation for the Ising Model (see Section 3.3). The left-hand figures depict the fraction of Susceptible/Infected/Recovered spins following each lattice sweep, while the right-hand figures illustrate the magnetization of the lattice throughout the sweeps. The three rows showcase three separate simulations, each using a different effective temperature.

Figure 12 illustrates the results of the simulations when the spins were allowed to become susceptible again after recovery (SIRS-model).

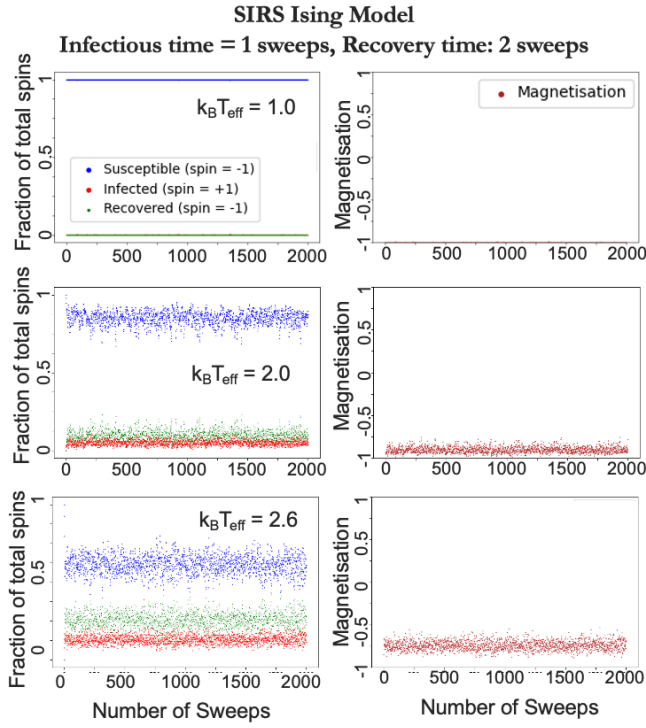


Figure 12: Plots illustrating the impact of incorporating infectious time and recovery into the epidemiological simulation for the Ising Model, and allowing recovered spins to become susceptible again (see Section 3.3). The left-hand figures depict the fraction of Susceptible/Infected/Recovered spins following each lattice sweep, while the right-hand figures illustrate the magnetization of the lattice throughout the sweeps. The three rows showcase three separate simulations, each using a different effective temperature.

Additional figures depicting the SIS, SIR, and SIRS simulations can be found in the appendix, showcasing variations in infectious and recovery times.

5. DISCUSSION

5.1 The standard Metropolis-Hastings procedure

The results of the standard Metropolis-Hastings procedure, shown in Figure 5., illustrate the expected behaviour of the Ising Lattice at various temperatures. Between temperatures $0-2 k_B T$, the absolute mean value of the lattice is $+1$, indicating an ordered phase. Between 2 and $3 k_B T$, $\langle |M| \rangle$ shows a continuous, smooth decrease, and after $k_B T=3$ it tends toward the disordered phase, $\langle |M| \rangle = 0$, indicating a second-order phase transition. In the phase transition region, it can be seen that the standard deviation increases, in line with the theory that in the critical region clusters form and the system has larger fluctuations. In Figure 6., it can be observed that the peak of $\Delta \langle |M| \rangle$ coincides for

both lattice sizes, successfully showing that the fluctuations in the system occur on all length scales at the critical region. The theoretical temperature is given by $k_B T_C = 2.27 J = 2.27$, according to Equation 7, which falls in line with the simulation's critical region of $2.2 < k_B T_C < 2.4$.

Figure 5. also shows the expected behaviour of the lattice energy. According to Equation 4, the lattice energy is a measure of the cluster formation within a lattice, indicating how much the spins are in parallel alignment (setting $J>0$). It can be observed that below the critical temperature, the lattice energy is at its minimum value of -2 , meaning that the whole lattice consists of one cluster. During the phase transition, smaller clusters start to form, hence E decreases. In the disordered phase, the spins tend to be in a random alignment, which results in E tending to a value of 0 .

5.2. Simulation of the Diffusion of COVID-19

Figure 7 provides a comparison of SN's work [1], and Figure 8 shows the report's attempt to reproduce the results are shown. Figure 9 depicts the attempt to improve the results, which were obtained by doing 10×2000 lattice sweeps (8×10^6 MC steps) instead of 1×30 sweeps (1.2×10^4 MC steps).

SN's methodology (Figure 7) shows that below $\sim 2.5^\circ\text{C}$ and above $\sim 35^\circ\text{C}$ the population is in a healthy state, while a disordered, epidemic phase is predicted between $\sim 2.5-35^\circ\text{C}$.

The comparison of Figures 7. and 8. provides an insight that the report's methodology produced a smoother and sharper phase transition, predicting the COVID-19 diffusion to occur between $5-31^\circ\text{C}$ instead of $2.5-35^\circ\text{C}$. By comparing Figures 8 and 9, it can be observed that the increase in the MC steps led to a sharper phase transition and reduced fluctuations, estimating the range of diffusion between $4-33^\circ\text{C}$. The increase in MC steps signifies a longer time for the algorithm to explore the configuration space, therefore it enhanced the probability of converging to the desired distribution values.

The bottom panels of Figures 7, 8, and 9 show the relationship between the energy of the lattice (interpreted as the mobility of the population by SN) and the magnetisation (health state). It can be seen that as the mobility increases, the diffusion of the disease also ascends. SN suggested that this correlation can be used as evidence that an increase in the mobility of the population results in a larger diffusion of COVID-19. While from an epidemiological perspective, this can be argued, from the mathematical perspective of the Ising Lattice the

correlation between $\langle M \rangle$ and $\langle E \rangle$ cannot be considered a causation. Firstly, this is because both the energy and the magnetisation of the lattice are calculated from the spin states, which depend on the temperature of the system. Secondly, there are configurations where the magnetism of the lattice is 0, which means an epidemic case, however, with different energies:

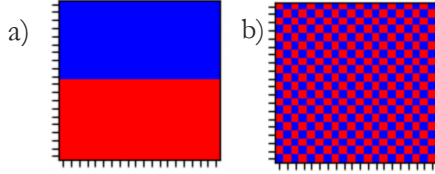


Figure 14: Two states of the Ising Lattice. The lattice in Figure a) has $M=0$ and $E = -1.8$, b) has $M=0$ and $E = 2.0$ with interaction strength $J=1$ and zero external magnetic field.

It has to be noted that configuration a) has a substantially lower entropy than configuration b) and is therefore unlikely. However, this example illustrates that the conclusion that an increase in mobility (lattice energy) increases the diffusion of COVID-19 (magnetisation) has to be more carefully considered.

SN's model [1] was developed during the COVID-19 pandemic 2020 for predictive purposes, therefore is interesting to compare the results in Figure 7. with now available empirical data of the pandemic. For this, Huang, Zhongwei, et al's paper, titled "Optimal temperature zone for the dispersal of COVID-19" [23] has been used.

They compared the number of confirmed COVID-19 cases between 21.01.2020 and 06.05.2020 with the daily air and dew-point temperatures, examining more than 3,750,000 COVID-19 cases from 185 countries/regions. Figure 15 shows that according to empirical data, the diffusion of COVID-19 is observed between 0 and 33°C. SN predicted this range to be between the range 2.5-35°C. There is a slight difference between the prediction and empirical data, however, the scale is appropriate. This difference can be due to not accounting for other important external factors of disease transmission, for example, the relative humidity (see Figure 15). Additionally, the difference between the lower-end prediction could be due to SN only considering positive temperatures, while COVID-19 has a small probability of diffusion below 0°C.

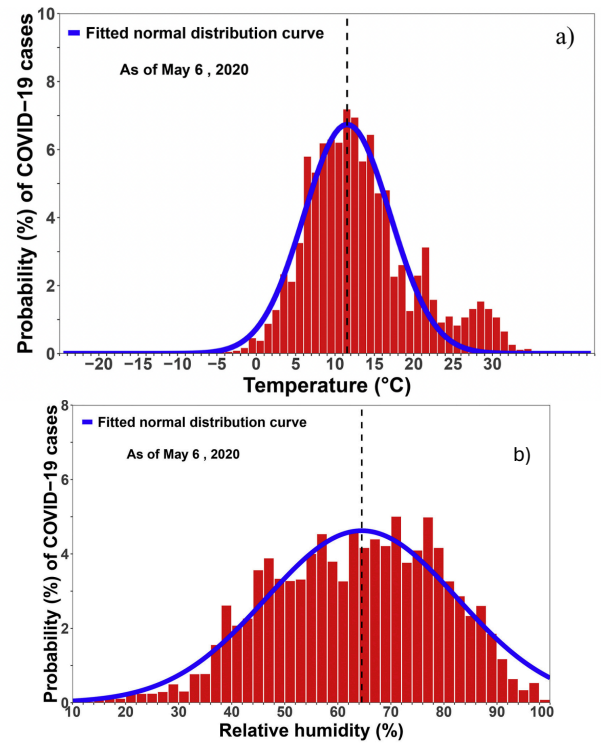


Figure 15: Zhongwei, et al's results [23] showing the relationship between daily confirmed cases of COVID-19 and temperature (a) or the relative humidity (b).

5.3. Extending the MCMC procedure to account for the duration of contagiousness and immunity

Figures 10, 11, and 12 show the outcomes of altering the duration for which spins remain in an infected or healthy state after infection. In this extension, the effective temperature serves as a measure of randomness, influencing the likelihood of infection occurrence and thereby impacting the transmission probability (as detailed in Section 3.3). The figures show that at a certain effective temperature (T_{eff}), the population of Susceptible/Recovered individuals begins to decline while the number of Infected individuals rises, and an epidemic occurs. This suggests that the effective temperature plays a crucial role in driving the increase in the number of infected spins. Consequently, this invites the interpretation of T_{eff} in real-life terms within the epidemiological analogy, that controls the outbreak of an epidemic. For instance, one could analogize temperature to factors such as mask usage or social distancing measures, higher temperatures corresponding to decreased effectiveness in containing the spread of infection.

It is noteworthy to observe that when the recovery time is set to 0 (as seen in Figure 10), introducing the infectious time results in the system fluctuating between fixed values, corresponding to specific configurations of the Ising Lattices. However, once the recovery time is introduced (as depicted in Figure 11), this discrete pattern of fluctuation is disrupted.

This behaviour can be attributed to the spins being kept infected for a certain number of sweeps with a 100% probability.

It is intriguing to note that the modified dynamical simulations exhibit similarities with the compartmental SIS, SIR and SIRS models, illustrated in Figure 16.

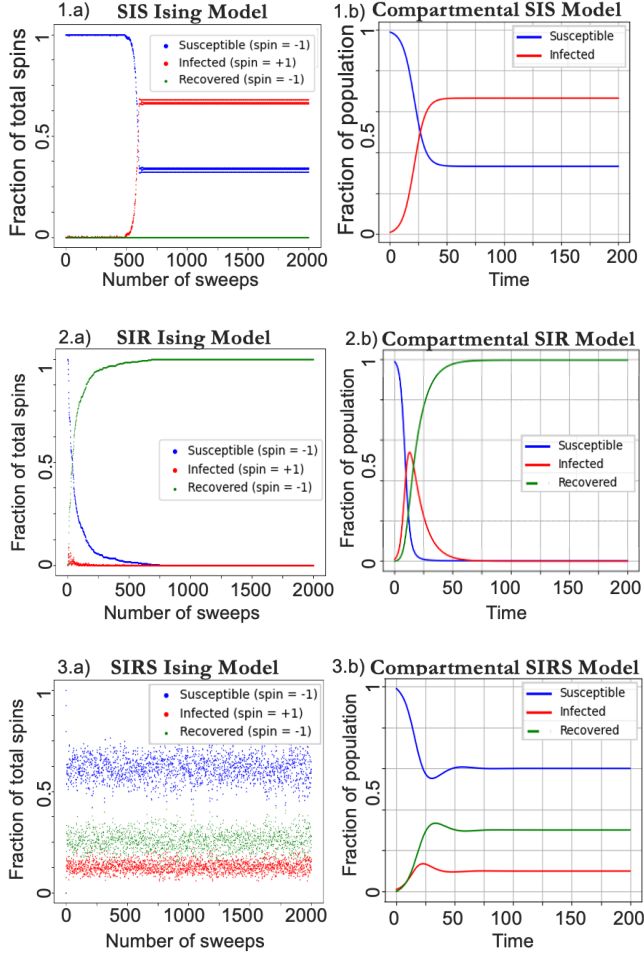


Figure 16: A comparison of specific epidemiology simulations of the Ising Model (plots to the left) and some example solutions of compartmental epidemiology models (to the right) (see Section 2.1). The epidemiology-Ising model simulation shown in panel 1.a) used the parameters of $T_{\text{eff}} = 0.8$, $I_t = 2$, $R_t = 0$, 2.a) employed $T_{\text{eff}} = 1.2$, $I_t = 2$, $R_t = \text{total}$, and 3.a) set $T_{\text{eff}} = 2.6$, $I_t = 1$, $R_t = 2$. The rates for the epidemiology ordinary differential equations were chosen manually to find similar behaviour for the Ising model plots. The rates were set to on panel 1.b) $\beta=0.3$, $\gamma=0.1$; 2.b) $\beta=0.6$, $\gamma=0.1$ and 3.b) $\beta=0.5$, $\gamma=0.3$, $\zeta=0.1$.

This comparison highlights that the extended MCMC method qualitatively aligns with epidemiological models, indicating a correspondence between the number of sweeps and the time elapsed in the differential equations. It also presents the opportunity that by adjusting the parameters of T_{eff} ,

the duration of infection and recovery time to real-life factors, the epidemiology extension of the Ising model can be utilized for simulating real-world disease dynamics. However, it is essential to note that such a comparison can only be made qualitatively, as the mathematical analogy between the SIS-SIR-SIRS differential equations and a dynamical simulation of the MCMC would have to be established first.

6. CONCLUSION

The implementation of the Monte Carlo Markov Chain method (MCMC), using the Metropolis-Hastings algorithm for a 2-dimensional Ising Lattice was successful, as a second-order phase transition was observed. Setting the interaction strength $J=1$, the external magnetic field to $h=0$ and using periodic boundary conditions, the critical temperature was estimated to be $2.2 < k_B T_C < 2.4$, in alignment with the theoretical value of $k_B T_C = 2.27$. T_C was obtained by examining the temperature range where the standard deviation of the magnetisation was the largest. The result could be improved by using more Monte Carlo steps and a more rigorous estimation of the critical temperature, for example using finite-size scaling. [24]

An analogy between the $+1/-1$ spin states within a ferromagnetic Ising Lattice and the infected/healthy individuals within a closed population has been established. The lattice magnetisation $M=-1$ has been interpreted as a healthy population, and $M=0$ as an epidemic case, with each spin having a 50-50% chance of being infected. Using this analogy, the diffusion of COVID-19 under lockdown conditions has been simulated successfully using SN's methodology [1]. The report's findings indicate that COVID-19 diffusion occurs between $5-30^\circ\text{C}$, slightly deviating from SN's result of $2.5-35^\circ\text{C}$. Despite this difference, the scale remains appropriate. In an effort to refine the results, the report increased the Monte Carlo steps from 1.2×10^4 to 8×10^6 , estimating the diffusion temperature range to be between $4-33^\circ\text{C}$. These results were compared to empirical data indicating COVID-19 diffusion between $0-34^\circ\text{C}$. Further enhancements could involve considering relative humidity, another factor influencing the scale of COVID-19 diffusion.

The MCMC method has been expanded to incorporate the duration of time an individual remains infected or immune after recovery. Temperature was considered as a measure of randomness, naming it as effective temperature, influencing the likelihood of infection occurrence. By integrating health history into the transition probability, it was argued that the successive lattice configurations could no longer be regarded as a

Markov Chain, leading to the interpretation of the Monte Carlo method as a dynamic calculation.

The Boltzmann weight was utilized as a probability for transitioning from a healthy to an infected state, although its justification was primarily qualitative. Consequently, the report could be extended by experimenting with different transition probabilities and comparing the results to real-world disease dynamics. Additionally, future studies could incorporate a probability for recovery post-infection, as currently, the recovery rate in the model was deterministic.

The modified Ising Model simulations yielded similar dynamics to the epidemiological differential equations describing the evolution of Susceptible-Infected-Recovered at certain effective temperatures, and a qualitative analogy between the number of Monte Carlo steps and time was drawn. However, this comparison remains inherently qualitative, and the epidemiological interpretation of temperature remains open. Therefore, it would be an intriguing prospect for expanding the project to explore a mathematical analogy between the two methods.

In conclusion, the primary objective of the project, which aimed to demonstrate the applicability of the Ising Lattice model in simulating disease diffusion within a closed population, has been effectively demonstrated.

7. APPENDIX

7.1 References

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7.2. Code for the MCMC extension

```
class IsingModel:

    def __init__(self, M, N, J, H):
        """
        Initialization.
        lattice is M by N sites
        J: interaction between neighbors (units: kT)
        H: background field (units: kT)
        """
        self.J = J # storing paramteres
        self.H = H
        self.M = M # row
        self.N = N # coloumn
        lattice_state = np.random.choice([-1, 1], size=(self.M, self.N)) # initialize lattice sites with -1 or 1 with equal prob
        self.lattice_state = lattice_state
        self.SIR_counter = np.ones((self.M, self.N)) * -1 # initiate an SIR_counter

    def update_SIR_counter(self, I_t, R_t, inf_i, inf_j):
        """
        After each lattice sweep, the SIR counter has to be updated.

        - I_t: - infectious time, the time (number of lattice sweeps / days) until the spin stays infectious (I_t > 0)
        - R_t: - recovery time, the time until the spin stays immune and in a healthy status (R_t > 0)

        - lattice_state: - the spin states of the Ising lattice, MxN matrix
                        - -1 represents HEALTHY state, +1 represents INFECTED state

        - SIR_counter: - the tracker for the Susceptible-Infected-Recovered status of the population, MxN matrix
                        - each [i, j] site corresponds to the SIR state of the lattice_state[i, j] spin
                        - values: x = -1 (SUSCEPTIBLE), the spin is susceptible to catch the disease
                                x = +1, +2, +3, ..., I_t (INFECTIOUS), the spin is infected since x sweeps (days)
                                x = -R_t, -(R_t-1), -(R_t-2), ..., -2 (RECOVERED), the spin remains healthy until the next x sweeps
                                x = 0 (RECOVERED), the spin remains healthy for all sweeps'''

        '1) Add +1 to all the values except for susceptible (-1) and immune spins (0)'
        self.SIR_counter[(self.SIR_counter != -1) & (self.SIR_counter != 0)] += 1

        '2) Keep track of the newly infected spins'
        self.SIR_counter[inf_i, inf_j] = +1

        '3) Recover those spins that have been infected longer than the infectious time'
        rec_i, rec_j = np.nonzero((self.SIR_counter > I_t))
        self.lattice_state[rec_i, rec_j] = - self.lattice_state[rec_i, rec_j] # flip the recovered spins
        if R_t > 0:
            self.SIR_counter[self.SIR_counter > I_t] = - R_t # indicate the recovery in the SIR_counter
        else:
            self.SIR_counter[self.SIR_counter > I_t] = 0 # indicate the inifinite immunity in the SIR_counter

    def flip_spin(self, i, j):
        """
        Flip the sign of spin at site (i, j)
        """
        self.lattice_state[i, j] = - self.lattice_state[i, j]

    def ferro_state(self, val):
        """
        Flip all spins into state = val
        """
        self.lattice_state = np.ones((self.M, self.N)) * val

    def E_spin(self, i, j):
        """
        Calculate energy of spin (i, j) with periodic boundary conditions
        """
        spin_here = self.lattice_state[i, j] # value of spin here

        if j == 0:
            spin_left = self.lattice_state[i, self.N - 1] # last column to the right
        else:
            spin_left = self.lattice_state[i, j - 1]

        if j == self.N - 1:
            spin_right = self.lattice_state[i, 0]
        else:
            spin_right = self.lattice_state[i, j + 1]

        if i == self.M - 1:
            spin_below = self.lattice_state[0, j]
        else:
            spin_below = self.lattice_state[i + 1, j]

        if i == 0:
            spin_above = self.lattice_state[self.M - 1, j]
        else:
            spin_above = self.lattice_state[i - 1, j]

        E = (- self.H*spin_here) - self.J*spin_here*(spin_above + spin_below + spin_left + spin_right)
        return E
```

```
def M_lattice(self):
    """
    Calculate net magnetization normalized by the number of spins
    """
    return 1.0 * np.sum(self.lattice_state) / (self.M * self.N)
```

```
def sweep_lattice(model, kT):
    """
    Perform N*M Monte Carlo spin flip attempts
    - model: Ising lattice model object instance
    - T: temperature
    - returns: the indices of the spins that have been flipped. Since the algorithm only evaluates
    susceptible, -1 spins, the flipped spins are the newly infected spins after the lattice sweep
    """
    sus_i, sus_j = np.nonzero((model.SIR_counter == -1)) # get the indices of the susceptible spins
    inf_i, inf_j = [], [] # store the indices of the spins that will get infected during the lattice sweep

    sus_seq = list(range(0, len(sus_i)))
    np.random.shuffle(sus_seq) # generate a random sequence of the susceptible spin indices

    for k in sus_seq:
        i = sus_i[k] # choose the susceptible spin (i,j)
        j = sus_j[k]

        E_old = model.E_spin(i, j)
        model.flip_spin(i, j) # flip the spin (infect the spin)
        E_new = model.E_spin(i, j)

        P = np.exp(-(E_new - E_old) / kT) # accept this proposal with probability P
        if np.random.uniform(0, 1) > P: # reject move, flip spin back to original state
            model.flip_spin(i, j)
        else:
            inf_i.append(sus_i[k])
            inf_j.append(sus_j[k]) # store the indices of the newly infected spins

    return inf_i, inf_j
```

```
def simulate_MCMC(M, N, J, H, kT, n_sweeps, n_burn_sweeps, I_t, R_t):
    """
    Monte Carlo simulation of Ising model in Canonical ensemble at kT, starting from a ferromagnetic case

    - M, N: dimensions of the 2D Ising lattice
    - n_sweeps: total number of Monte Carlo sweeps to perform in the simulation
    - n_burn_sweeps: number of sweeps for equilibration (no samples taken during this period)
    - I_t, R_t: infectious time, recovery time (number of lattice sweeps a spin stays infected / recovered)
    """
    # --- INITIATIONS ----
    t0 = time.time() # keep track of simulation time
    k = 0 # number of sampling sweep rounds
    S = [] # initiate arrays to store the total number of SIR compartments
    I = []
    R = []
    M_samples = np.zeros(n_sweeps) # initiate arrays that will store the macrostate values, Mr and Er for each sweep
    E_samples = np.zeros(n_sweeps)
    E_samples_eq = np.zeros(int(n_sweeps - n_burn_sweeps))
    M_samples_eq = np.zeros(int(n_sweeps - n_burn_sweeps))
    spin_tracker = np.zeros([n_sweeps, 2]) # keep track of the health and SIR value of a single spin for control purposes

    model = IsingModel(M, N, J, H) # initiate the Ising lattice
    model.ferro_state(-1) # start from all spin HEALTHY state (s[i,j] = -1)

    # --- MONTE CARLO SAMPLING - FINITE IMMUNITY ----
    for n in range(n_sweeps):
        I.append(np.count_nonzero(model.SIR_counter >= 1)) # keep track of the total values of the SIR compartments
        S.append(np.count_nonzero(model.SIR_counter == -1))
        R.append(np.count_nonzero((model.SIR_counter < -1) | (model.SIR_counter == 0)))

        inf_i, inf_j = sweep_lattice(model, kT) # 1) sweep the lattice, infect the susceptible spins
        model.update_SIR_counter(I_t, R_t, inf_i, inf_j) # 2) update the SIR_counter based on the newly infected spins
        M_samples[n] = model.M_lattice()
        E_samples[n] = model.E_lattice()
        spin_tracker[n,0] = model.lattice_state[9,9]
        spin_tracker[n,1] = model.SIR_counter[9,9]

        if n >= n_burn_sweeps: # the system equilibrated, start taking equilibrium samples
            E_samples_eq[k] = model.E_lattice()
            M_samples_eq[k] = model.M_lattice()
            k += 1

    return spin_tracker, E_samples, M_samples, E_samples_eq, M_samples_eq, np.array(S), np.array(I), np.array(R)
```

```

def repeated_simulation(M, N, J, H, I_t, R_t, kT_list, n_repeat, n_sweeps, n_burn_sweeps):
    """
    - Repeats the Monte Carlo sampling at temperatures kT in kT_list, n_repeat times
    - Returns: xy_arr[i][j][k] arrays, where i is the number of repetition of the MCMCM simulation (n_repeat),
              j is the index of the temperature of the simulation (kT_list[j]), and k is the number of a sweep
    """

    # --- INITIATIONS ---
    E_arr = np.zeros([n_repeat, len(kT_list), n_sweeps]) # arrays for Er and Mr samples at each sweep
    M_arr = np.zeros([n_repeat, len(kT_list), n_sweeps])
    E_eq_arr = np.zeros([n_repeat, len(kT_list), n_burn_sweeps]) # arrays for equilibrium samples
    M_eq_arr = np.zeros([n_repeat, len(kT_list), n_burn_sweeps])
    S_arr = np.zeros([n_repeat, len(kT_list), n_sweeps]) # arrays to keep track of the SIR compartments
    I_arr = np.zeros([n_repeat, len(kT_list), n_sweeps])
    R_arr = np.zeros([n_repeat, len(kT_list), n_sweeps])
    spin_tracker_arr = np.zeros([n_repeat, len(kT_list), n_sweeps, 2]) # array to keep track of the health and
                                                                    # the SIR status of a single spin for control purposes

    # --- REPEATED MCMC SIMULATION ROUNDS ---
    print(f'Starting simulation with I_t={I_t}, R_t={R_t}', flush = True)
    for i in range(n_repeat):
        print(f'Sampling round {i+1}...\n', flush=True)

        # --- SIMULATE THE SWEEPS AT VARIOUS kT ---
        for j in range(0, len(kT_list)):

            t0 = time.time()
            spin_tracker, E_samples, M_samples, E_samples_eq, M_samples_eq, NI, S, I, R = simulate_MCMC(M, N, J, H, kT_list[j],
                                                                                                     n_sweeps, n_burn_sweeps,
                                                                                                     I_t, R_t)

            E_arr[i,j] = E_samples
            M_arr[i,j] = M_samples
            E_eq_arr[i,j] = E_samples_eq
            M_eq_arr[i,j] = M_samples_eq
            S_arr[i,j] = S
            I_arr[i,j] = I
            R_arr[i,j] = R
            spin_tracker_arr[i, j] = spin_tracker

            print(f'    kT = {round(kT_list[j], 2)}, M_mean = {np.mean(M_samples_eq)}, S_mean = {np.mean(S)},
                  I_mean = {np.mean(I)}, Sim time: {round(time.time()-t0, 4)} sec')

        print(f'Finished sampling round {i+1}.\n', flush=True)
    print(f'Finished simulation.', flush=True)

    return spin_tracker_arr, E_arr, M_arr, E_eq_arr, M_eq_arr, S_arr, I_arr, R_arr

```

7.3. Additional results for the MCMC extension

